

BCD Arithmetic

Assuming a 4-bit adder primitive

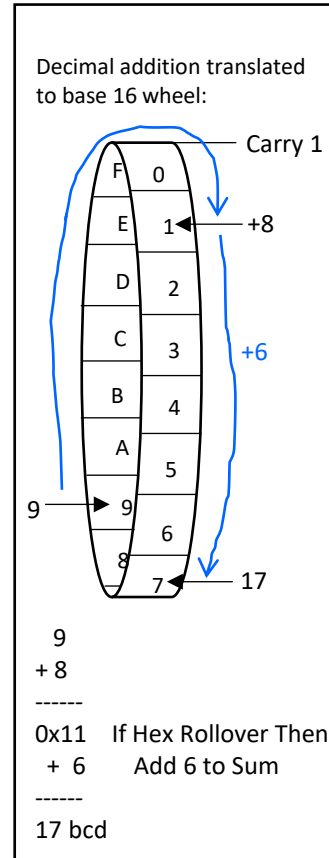
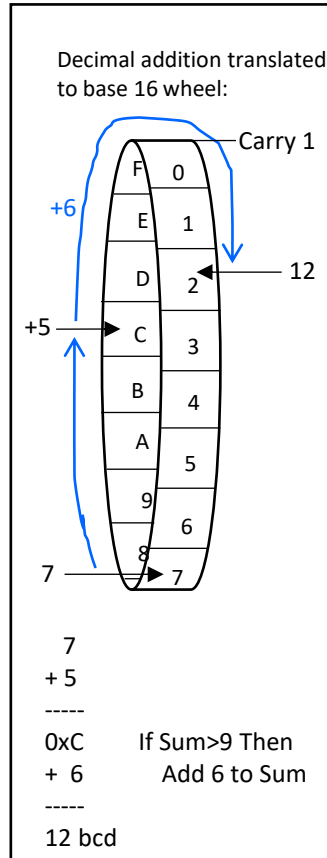
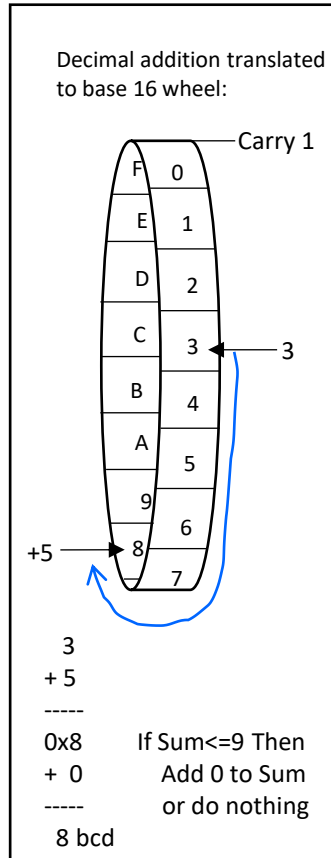
e.g. 7483, 74283, etc.

BCD Addition

Given a 4-bit adder:

$$A_{10} + B_{10} + C_{in} \rightarrow (A_{BCD} + B_{BCD} + C_{in}) \bmod 16 = S_{16},$$

$$[(A_{BCD} + B_{BCD} + C_{in}) \bmod 16 \geq 16] = Co$$



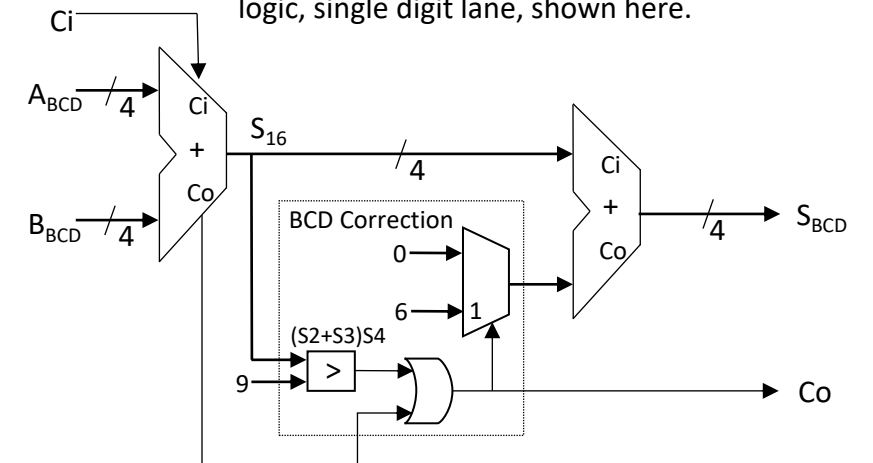
(Op1+Op2+Ci) ₁₀	Sum ₁₆	Co	Sum ₁₀	Co
0	0	0	0	0
1	1	0	1	0
2	2	0	2	0
3	3	0	3	0
4	4	0	4	0
5	5	0	5	0
6	6	0	6	0
7	7	0	7	0
8	8	0	8	0
9	9	0	9	0
10	A	0	0	1
11	B	0	1	1
12	C	0	2	1
13	D	0	3	1
14	E	0	4	1
15	F	0	5	1
16	0	1	6	1
17	1	1	7	1
18	2	1	8	1
19	3	1	9	1

Sum of 2, 4-digit numbers

	Co	Ci	Co	Ci	Co	Ci	Co	
7693		0111	1		0110	1		1001
+9684		+1001			+0110			+1000
17377	1	0001			1101	1		0001
		+0110			+0110			+0110
								+0000
	1	0111	1	0011		0111		0111

Note: Carry out of either the main summer or the BCD corrector is a BCD carry out.

All of which leads to the idea of the BCD adder logic, single digit lane, shown here.



Number Complements

Radix complements and their sidekicks, diminished radix complements are a numeric tool allowing subtraction be performed by addition.

- A number N , in radix R has complement numbers.
- In particular a number N in radix R has its R 's complement and its diminished R 's complement, or $(R-1)$'s complement.
- The R 's complement is the $(R-1)$'s complement plus 1.
- The $(R-1)$'s complement is computed by taking each digit of N and subtracting it from $R-1$.

EX: If $R=2$ and $N=1110$, the 1's complement is 0001,
and the 2's complement is 0001 plus 1 or 0010.

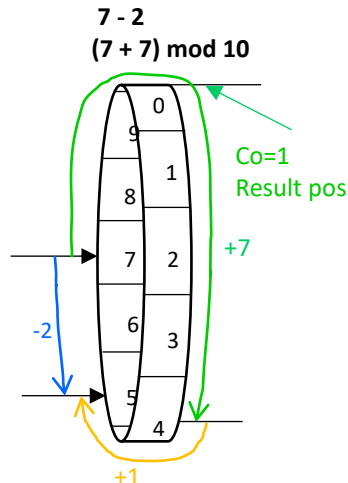
9's and 10's Complement Subtraction

To take the 9's complement of a number, say 752, subtract each digit of the number from 9.

$$\begin{array}{r} 999 \\ - 752 \\ \hline 247 \end{array}$$

To subtract two numbers using 9's complement arithmetic:

- Add the 9's complement of the subtrahend to the minuend.
- If there is a carry out of the most significant digit, then add 1 to the result. The result is positive.
- If there is no carry out of the most significant digit, the result is negative and must be 9's complemented.



$$5841 - 2963 = 2878$$

$$\begin{array}{r} 9999 \\ - 2963 \\ \hline 7036 \\ + 5841 \\ \hline 1 \mid 2877 \\ 1 \\ \hline 2878 \end{array}$$

9's complement of subtrahend

Add the minuend

Add MSD carry to sum

Difference is positive

$$4691 - 6849 = -2158$$

$$\begin{array}{r} 9999 \\ - 6849 \\ \hline 3150 \\ + 4691 \\ \hline 0 \mid 7841 \\ 9999 \\ - 7841 \\ \hline - 2158 \end{array}$$

9's complement of subtrahend

Add the minuend

Difference is negative

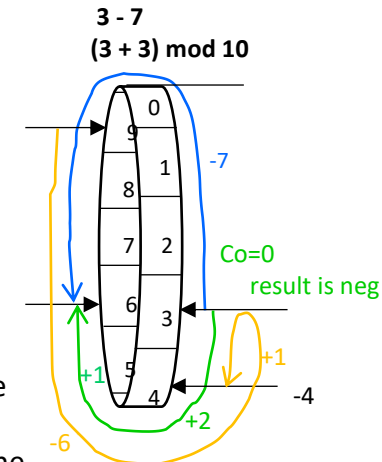
9's complement of difference

To take the 10's complement of a number, say 750, subtract each digit of the number from 9 and add 1 to the result.

$$\begin{array}{r} 999 \\ - 750 \\ \hline 249 \\ + 1 \\ \hline 250 \end{array}$$

To subtract two numbers using 10's complement arithmetic:

- Add the 10's complement of the subtrahend to the minuend.
- If there is a carry out of the most significant digit, the result is positive.
- If there is no carry out of the most significant digit, the result is negative and must be 10's complemented.



$$5841 - 2963 = 2878$$

$$\begin{array}{r} 9999 \\ - 2963 \\ \hline 7036 \\ + 1 \\ \hline 7037 \\ + 5841 \\ \hline 1 \mid 2878 \end{array}$$

10's complement of subtrahend

Add the minuend

Difference is positive

$$4691 - 6849 = -2158$$

$$\begin{array}{r} 9999 \\ - 6849 \\ \hline 3150 \\ + 1 \\ \hline 3151 \\ + 4691 \\ \hline 0 \mid 7842 \\ 9999 \\ - 7842 \\ \hline 2157 \\ + 1 \\ \hline - 2157 \end{array}$$

10's complement of subtrahend

Add the minuend

Difference is negative

10's comp of diff

10's Complement Arithmetic With A 4-Bit Binary Adder

For a 4-bit binary number, or hex digit H, the F's complement of H is $0xF - H$.

*Note that the F's complement of a 4-bit number is also the 1's complement of that 4-bit number.

For a BCD digit N, the 9's complement of N is the F's complement of N minus the number of hex digits not part of the decimal digit set (A,B,C,D,E,F), which is:

$$0xF - N - 0x6,$$

$$\text{Or more specifically, } (0xF - N - 0x6) \bmod 16.$$

And that is the same as:

$$(0xF - N + 0xA) \bmod 16$$

And

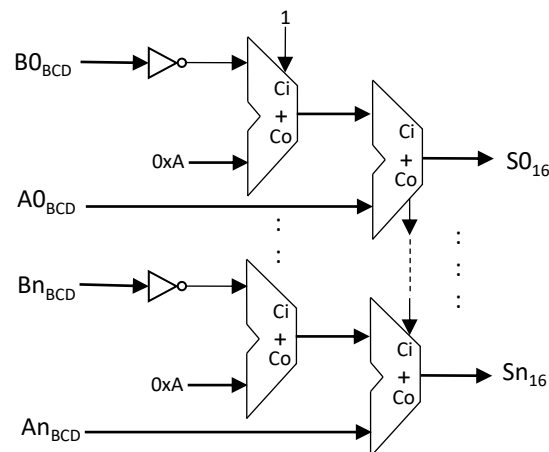
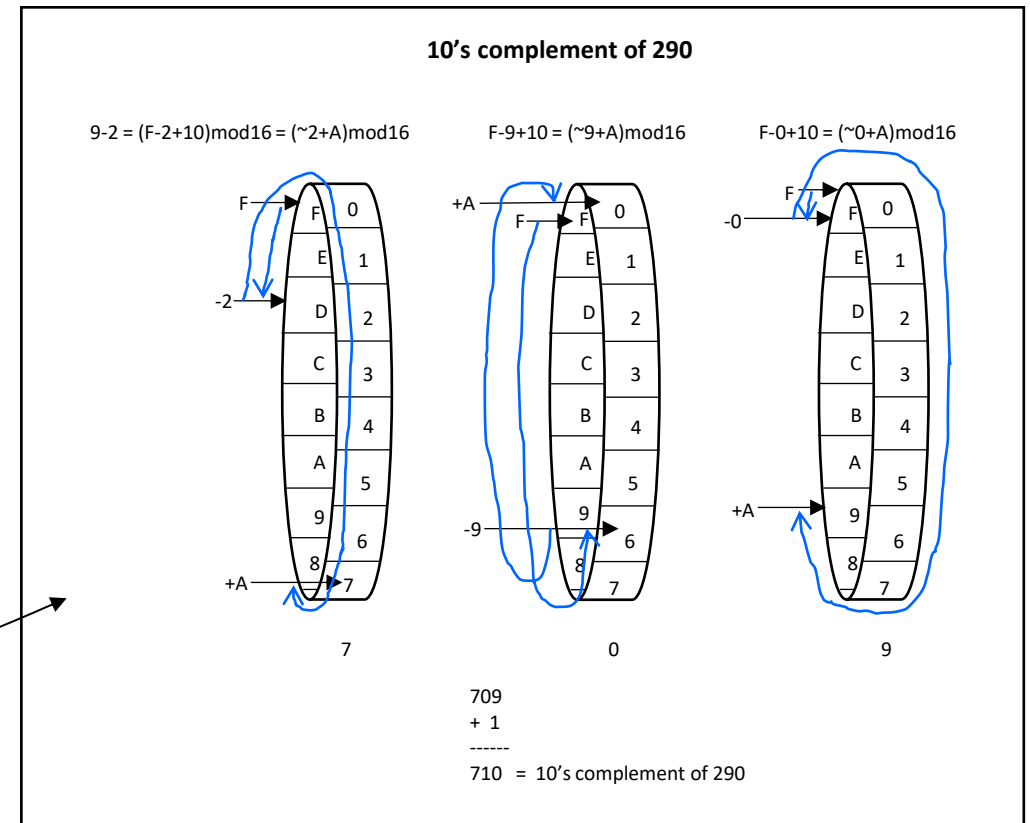
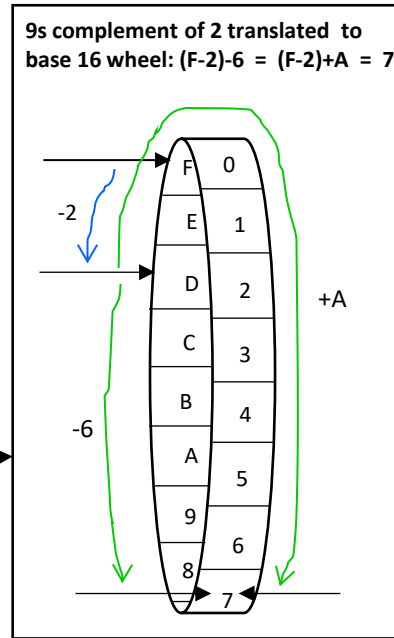
$$(*1'scomp(N) + 0xA) \bmod 16$$

Which is

$$\sim N + 0xA \leftarrow \text{9's complement of N}$$

The 10's complement of a number N is still just the 9's complement plus 1.

This suggests the idea of a BCD subtractor looks in part like the diagram on the right.



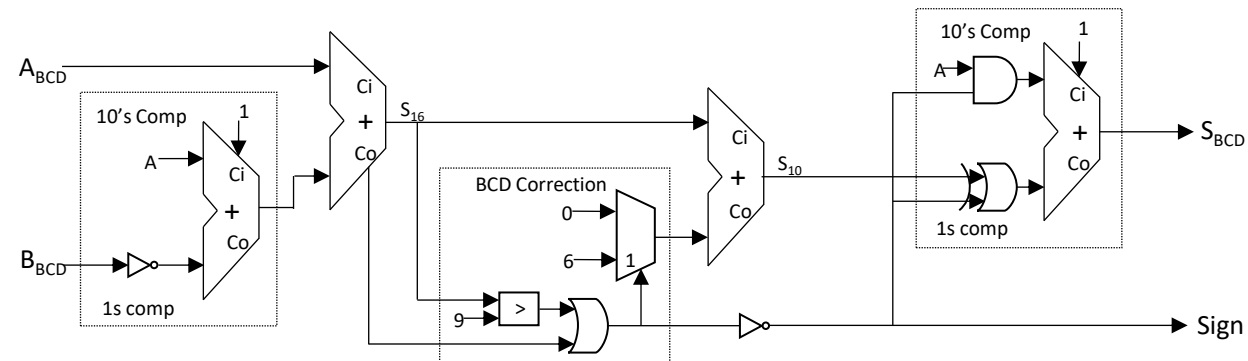
BCD Subtraction

		Co		Ci	Co		Ci			
A	47		0100			0111		minuend		
B	-28		-0010			-1000		subtrahend		
	19									
			1101			0111		1's complement of subtrahend		
			+1010			+1010	1	add 10 (and 1 more for LSD)		
		1	0111		1	0010		10's complement of subtrahend		
			+0100			+0111		add minuend		
			1011			1001		difference		
			+0110			+0000		bcd correction		
		1	0001			1001				
		Final carry=1 difference is pos (A>=B)								

		Co		Ci	Co		Ci			
A	28		0010			1000		minuend		
B	-41		-0100			-0001		subtrahend		
	-13									
			1011			1110		1's complement of subtrahend		
			+1010			+1010	1	add 10 (and one more)		
		1	0101		1	1001		10's complement of subtrahend		
			+0010	1		+1000		add minuend		
		0	1000		1	0001		difference		
			+0000			+0110		bcd correction		
			1000			0111				
		Final carry=0 difference is neg (A<B)								
			0111			1000		1's complement		
			+1010			+1010	1	add 10 (and one more)		
			0001		1	0011		10's complement		

Op1-Op2	Co	A + 10#(B)	SUM
-9	0	1	-9
-8	0	2	-8
-7	0	3	-7
-6	0	4	-6
-5	0	5	-5
-4	0	6	-4
-3	0	7	-3
-2	0	8	-2
-1	0	9	-1
0	0	A	0
1	0	B	1
2	0	C	2
3	0	D	3
4	0	E	4
5	0	F	5
6	1	0	6
7	1	1	7
8	1	2	8
9	1	3	9

A single digit BCD subtractor conceptually looks like this



Example of Combinational Three-Digit BCD Subtractor

